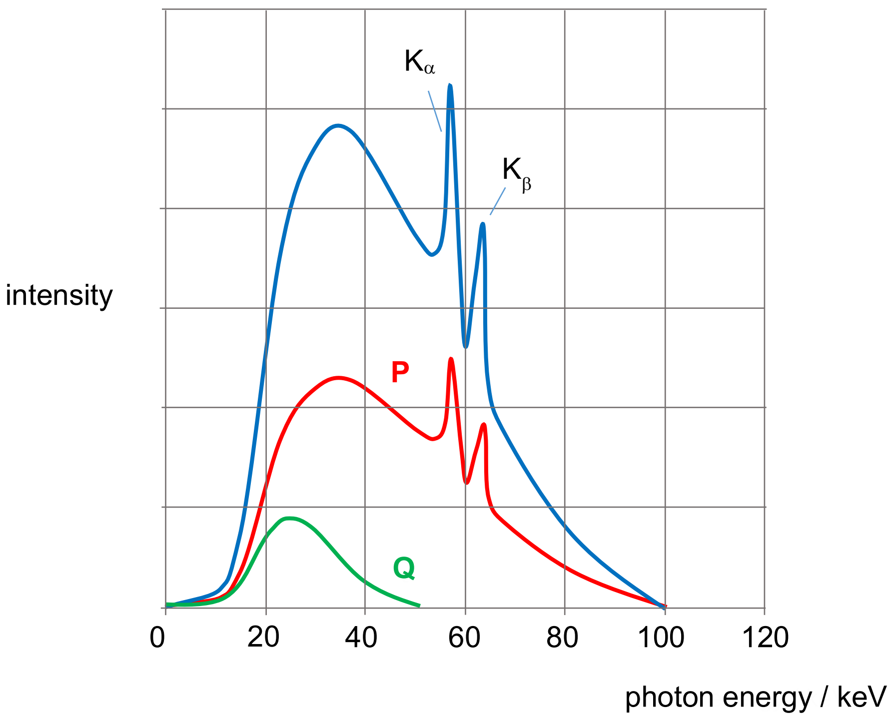
Qns	
6(a)(i)	V = 100 kV. maximum photon energy is all 100 keV op KE lost by a single bombarding electron having accelerated across a potential difference of 100 kV

6(a)(ii)



6(a)(iii) K-shell electron in target atom ionised/knocked out by high energy incident electron, creating a vacancy/ hole.

L-shell electron in target atom drops to fill up K-shell vacancy.

Energy difference given out as K_α photon.

6(b)

$$\begin{aligned}\text{L-shell ionisation energy} &= (\text{K}_\alpha \text{ photon energy}) + (\text{K-shell ionisation energy}) \\ &= 59.3 + (-69.5) \\ &= -10.2 \text{ keV}\end{aligned}$$

6(c) Any 1 of the following:

- High melting point – since large amount of heat generated in target metal
- Large specific heat capacity – lower temperature rise for given amount of heat absorbed
- Good thermal conductivity – conducts heat away

7(a) the value of the direct current
that dissipates thermal energy at the same rate in a resistor